One internal target, two contribution coordinates

Choose a block/head and define the scalar before computing either explanation.

$$g(z) = \sum_{j=0}^{196} A(x)_{0j} V_j(z), \qquad f(z) = u^\top g(z)$$

$$u = \frac{g(x) - g(x_0)}{\|g(x) - g(x_0)\|_2}$$

A(x): clean CLS attention; A and u stay fixed for this explanation.

Value features V(z) vary with the input.

a NNS · input-space integrated gradients

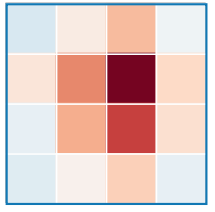


Reference x_0 | Path input $x\alpha$ | Clean image x

$$x_\alpha = x_0 + \alpha(x - x_0)$$

$$R_i = (x_i - x_{0i}) \int_0^1 \partial_i f(x_\alpha)\, d\alpha$$

Sum RGB channels with sign, then sum into fine cells.



Input-coordinate scores

Within-patch values can differ.

Computed from f, x and x_0.

Blue: negative · red: positive

b Matched token baseline

Evaluate clean and reference Value features

$$c_j = A(x)_{0j}\, u^\top [V_j(x) - V_j(x_0)]$$

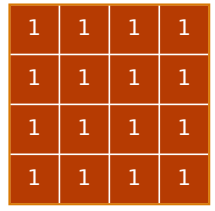
Attention weight × target-directed Value change

196 patch-token scalars

CLS scalar kept separately

Example token A

$$c_A = 16$$



One patch

16 fine cells

$c_j/16$ per fine cell

Reference x_0: normalized zero / RGB 0.5. Diagrams and numbers are illustrative.

Deep tokens mix spatial information: input-patch attribution sums need not equal the indexed token contributions.